

Analytical and numerical modelling of wave propagation in (meta)poro(elastic) layers

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This chapter lays the foundation for the thesis by providing the fundamental theoretical notions and methods used in this work, in order to guide the reader through the various chapters that will follow. At first, an overview of the acoustic modelling of porous media is proposed. Following the presentation of the state of art in the introduction and of the areas of focus that have been identified from the latter, the key notions and methods for the computation of the scattering of waves by metaporous structures will be presented. From there, the formalism for the calculation of dispersion curves will be presented along with an overview of the methods used for such computations. Both of these topics will be illustrated with preliminary results in simple configurations.

1.1 | Wave propagation in homogeneous layers

In the previous section, a formulation for rigid-frame porous media and two formulations for poroelastic media modelling have been presented. Concluding the presentation of these motion equations, some characteristics of the wave propagation in unbounded porous and poroelastic media have been shown. The present section proposes the introduction of a geometry, that is to say boundaries or interface conditions delimiting different domains characterized each by a different type of material. Generally, the configurations studied in this work assume a monochromatic plane wave originated from a fluid half-space, i.e. air, impinging at the surface of a structure and is then scattered (reflected and transmitted) from and through each interface.

If the material in which the wave propagates is bounded by two parallel, free, and flat surfaces, i.e., a plate, then it is referred to as a layer. The waves supported by the medium propagate by reflecting alternatively from one surface to the other; they are said to be guided along the layer. These observations of successive reflections and refractions lead to the concept of waveguides, but their analytical descriptions often leads to writing complex system that can be difficult to solve. To determine the modes of a structure, one must seek to solve the propagation equations and the boundary conditions imposed at its boundaries. In a first section, the scattering of a plane wave

by a homogeneous porous layer is investigated along with the various method available. Next, the methods for the computation of the modes of these configurations are presented. In a subsequent section, more complex geometries, those of metaporous structures, are of interest and both methods for the computation of the scattering coefficients and the modes of those configurations will be discussed.

1.1.1. Analytical plane wave modelling

Modeling acoustic propagation in homogeneous layers by assuming plane waves is the simplest formalism in acoustics. We assume these plane waves such that they satisfy the motion equations, e.g. the ones introduced in ?? or ??. Introducing a simple problem, that of the scattering of an incident plane by a porous layer, we present an example of plane wave modelling for a simple system. The system is 2D with coordinates $\mathbf{x} = (x_1, x_2)$ constituted of a porous layer with thickness h and thus two interfaces of infinite extent along x_1 , there is a fluid-porous interface at $x_1 = h$ and a porous-fluid interface at $x_1 = 0$. Both media of the air half-spaces and porous layer are modeled as fluid, thus the interfaces conditions should impose continuity of the pressure fields and normal particle displacements.

Assuming a $e^{i\omega t}$ time-dependance, the incident pressure field originated from the upper domain that satisfy the Helmholtz equation takes the form of,

$$p_{0+} = a_+ e^{-ik_1 x_1} e^{-ik_{2,0}(x_2-h)} + a_- e^{-ik_1 x_1} e^{ik_{2,0}(x_2-h)}. \quad (1.1)$$

The pressure field in the porous layer is expressed as a sum of forward and backward waves,

$$p_p = b_+ e^{-ik_1 x_1} e^{-ik_{2,eq} x_2} + b_- e^{-ik_1 x_1} e^{ik_2(x_2-h)}, \quad (1.2)$$

and that in the lower air domain is,

$$p_{0-} = c e^{-ik_1 x_1} e^{ik_{2,0} x_2}, \quad (1.3)$$

with wavenumbers such that they hold for the Snell-Descartes law that states $k_1^2 + k_{2,0}^2 = k_0^2$ and $k_1^2 + k_{2,eq}^2 = k_{eq}^2$. The wavenumber should be imposed with $k_1 = k_0 \sin \theta$, $k_2 = -k_0 \cos \theta$, with θ the angle of incidence, that governs the direction of propagation of the incident field. The displacement fields are additionally expressed using the momentum equation relating the latter with the pressure fields as $\partial_2 p = -i\omega \rho_i u_2$, $i = (0, eq)$.

Next, the interface conditions describe the relationships between properties at the boundary between two media, including the continuity or vanishing of given fields. If a media is terminated by a surface (e.g. a rigid wall), a constraint should be applied on the fields to represent it, denoted as boundary conditions. Assuming a continuity between the two fluids in the present case with interfaces located at $x_2 = 0$ and $x_2 = h$, we obtain the following conditions,

$$p^t(h) = p^l(h), \quad u_2^t(h) = u_2^l(h) \quad p^l(0) = p^b(0) \quad u_2^l(0) = u_2^b(0), \quad (1.4)$$

with superscripts b and t denoting respectively the bottom and top interfaces.

The full analytical description of a system constituted from one or several interfaces can be done by writing a linear system built from the interface conditions, which vector of unknowns are the amplitudes of each of the existing waves in the system, that is $(b_+ \ b_- \ a_+ \ c)^T$. The forcing right-hand side corresponds to the incident wave terms. The solution of the system yields all unknown amplitudes of the system, in particular the reflection and transmission coefficients. Imposing the

interface conditions leads to writing the following matrix system,

$$\begin{pmatrix} 1 & e^{ik_{2,eq}h} & 0 & -1 \\ \frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,eq}}{\rho_{eq}}e^{ik_{2,eq}h} & 0 & \frac{k_{2,0}}{\rho_0} \\ e^{ik_{2,eq}h} & 1 & -1 & 0 \\ \frac{k_{2,eq}}{\rho_{eq}}e^{ik_{2,eq}h} & -\frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,0}}{\rho_0} & 0 \end{pmatrix} \begin{pmatrix} b_+ \\ b_- \\ a_+ \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ a_- \\ \frac{k_{2,eq}}{\rho_{eq}}a_- \end{pmatrix}. \quad (1.5)$$

From this expression, the system can simply be solved to obtain all scattered amplitudes b_+ , b_- , a_+ and c . In the case of a rigidly-backed layer, the interface at $x_2 = h$ is replaced by a rigid termination. For (equivalent) fluids, a hard wall or Neumann-type boundary condition is translated by a vanishing normal particle displacement field along the boundary, that is $u_2(\Gamma) = 0$. The term depending on c thus vanishes, since p_{0-} does not exist; this reduces the size of the system. In this case, the reflection coefficient R corresponds to the amplitude a_+ .

This method is denoted in this work as the Global Method (GM), since all of the available data is used in the formulation of the problem as the unknowns are all wave amplitudes. As described in Section ?? of Chapter ??, a general formalism can be used for this method and any stack of any layer can be described with the GM, by putting together a matrix system that contains all interface conditions in the system, in the general form,

$$\mathbf{M}(\omega)\mathbf{x} = \mathbf{f}, \quad (1.6)$$

where $\mathbf{M}$ corresponds to the interface conditions involved in the system, $\mathbf{x}$ contain the wave amplitudes and $\mathbf{f}$ is the forcing term, the incident plane wave amplitudes.

A few results for the simple cases of the scattering of a plane wave by a rigidly-backed layer and in the case of a symmetric configuration (with air coupling at both interfaces) are represented by the diagrams in Figs 1.1a and b and results in Figs 1.1c and d, respectively. The GM results are presented by the continuous lines. There, the layer for both porous and PEM are both $h = 5$ cm thick and the incident wave propagates with a $\theta = 30^\circ$ angle. The parameters for each material corresponds to the one used in Table ?? from Chapter ?. The system used to obtain the reflection coefficient for the PEM layer is done in a similar fashion as the equivalent fluid and can be read in the appendices of Ref.¹.

The first case, also referenced to as a reflection problem compares the equivalent fluid and the full Biot model response. The two curves follow the same trend but additional effects due to the skeleton motion can be observed on the continuous lines while the equivalent fluid response shows to be smoother. For all configurations, the melamine layer is acoustically transparent at lower frequency and absorption can be noticed at slightly higher frequencies.

1.1.2. Transfer Matrix Method

The widely used method we know today as the Transfer Matrix Method (TMM), was initially developed by the works of Thomson² and Haskell³ in order to describe elastic wave propagation in multilayer systems. This method can be applied to a range of wave physics among geophysics, quantum mechanics, electromagnetics and obviously acoustics. TMM allows for an easy multiphysics coupling with interface matrices; its core lies in the evaluation of the state vector that regroups the variables of interest in the system and evaluate them at each interface, which is possible thanks to the interface conditions. Then the physical field amplitudes stored in the state vector are propagated from one layer to the next. The transfer matrix $\mathbf{T}$ relating the pressure and normal displacement fields from the bottom (located at $x_2 = 0$) to the top interface (located at $x_2 = h$) of a fluid layer is thus,⁴

$$\underbrace{\begin{pmatrix} p(0) \\ u_2(0) \end{pmatrix} = \begin{pmatrix} \cos(k_{eq}b) & iZ_{eq} \sin(k_{eq}b) \\ iZ_{eq}^{-1} \sin(k_{eq}b) & \cos(k_{eq}b) \end{pmatrix} \begin{pmatrix} p(b) \\ u_2(b) \end{pmatrix}}_T, \quad (1.7)$$

where k_{eq} is the bulk wavenumber of the wave propagating in the equivalent fluid and Z_{eq} is the medium impedances. The extension to PEM makes of a state vector that includes the fields, $(\hat{\sigma}_{12} \ u_2^s \ u_2^t \ \hat{\sigma}_{22} \ p \ u_1^s)^T$. More details are given in Chapter 11 of Allard and Atalla textbook⁵.

For stacks of layers, an amount of N layers of the same nature are easy to manipulate since $T_{\text{global}} = \prod_{n=0}^N T_n$ since they have the same size. multiphysics with interface matrix that relate the relation between the various fields at an interface (non square matrices). This method was deep dived in several textbooks among which we propose Refs.^{5,6} for additional details. The TMM also allows for the computation of complex system by an assembly of elements including admittance matrices in order to consider Helmholtz Resonator (HR) embedded in an equivalent fluid layer⁷.

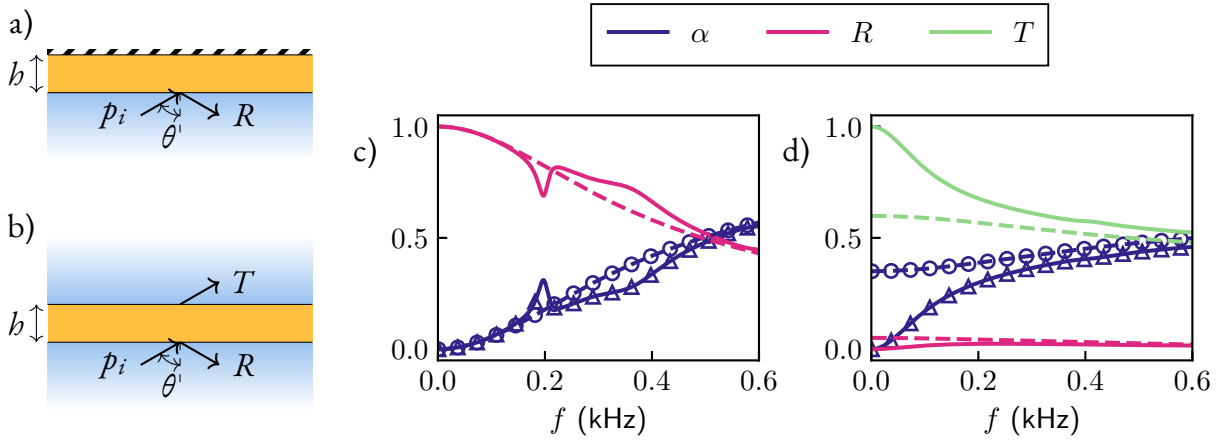


Fig. 1.1: Diagram for the a) reflection, b) transmission problem. Scattering coefficients from a poroelastic layer (continuous) and equivalent fluid layer (dashed) computed using the GM for the c) reflection, d) transmission problem. Results computed with the TMM are also shown for the absorption coefficient for the equivalent fluid (o) and for the absorption coefficient (Δ)

While very versatile, it remains unstable in limit cases at high frequencies or for large thickness of the layers. While the formulation is mathematically exact, the evaluation of the system involves finite-precision calculation which leads the very sensitive exponential terms to diverge in the previously specified regimes. From the evaluation of the transfer matrix of the global system, the scattering matrix can be obtained and thus the reflection and transmission coefficients by simply manipulating the expressions for the fields contained in the state vector with the scattering coefficients.

Considering a rigidly-backed layer, and thus $u_2(b) = 0$, we obtain a reflection coefficient,

$$R_b = \frac{T_{11} - Z_{eq}T_{21}}{T_{11} + Z_{eq}T_{21}}. \quad (1.8)$$

In the case of the symmetric layer, the reflection and transmission coefficients are determined from the scattering matrix as well,

$$T_s = \frac{e^{-ikL}}{T_{11} + T_{12}/Z_{eq} + Z_{eq}T_{21} + T_{22}}, \quad R_s = \frac{T_{12}/Z_{eq} - Z_{eq}T_{21}}{T_{11} + T_{12}/Z_{eq} + Z_{eq}T_{21} + T_{22}}. \quad (1.9)$$

which is only valid for reciprocal, time-invariant and linear assumptions. The absorption coefficient for the same two configurations are calculated as $\alpha_b = 1 - |R_b|^2$ and $\alpha_s = 1 - |R_s|^2 - |T_s|^2$ and are shown in Figs 1.1c and d with markers. It can be seen that the same results (up to the respective numerical precision of each method) are obtained.

1.2 | Guided waves for dispersion problems

In this section, the solutions for free systems, that is without any excitation or forcing term is of interest. The form of the modes of wave propagation in layer is discussed and various methods for the calculation of these modes are presented.

The existence of an interface in the geometry leads to the propagation of not only the bulk waves but also of so-called surface waves⁸ where the interface itself acts as the support for propagation. Different types of waves propagate along the interface. An example can be surface water waves that follow the interface between air and water; all the energy is localized at the interface. A similar behaviour exist for seismic waves where the interface of a solid substrate supports the wave propagation, that is Rayleigh (longitudinal motion) and Love waves (transverse motion) which both show an energy decaying with depth. Scholte waves are waves that propagate along the interface between a fluid and a solid medium. In fluid layers, the propagating waves are similar to studying a shear horizontal (SH) wave in an elastic layer. One mode propagates at zero frequency, while countless other modes propagate with a cut-off frequency that corresponds to the successive quarter-wavelength resonances, essentially depending on the layer thickness and the nature of the boundary/interface conditions.

From the previously obtained linear systems in Section 1.1.1 for the configurations, and removing the incident wave from the system, i.e. the forcing (right-hand side) term, an homogeneous linear system is obtained, representing the guiding of waves in the layer. This set of equations becomes singular when its determinant vanishes for a given parameter, allowing to identify the modes of the structure. Several formalism can be adopted: solving for frequencies while imposing the wavenumber or the other way around. The first one is useful when interested in the time-dependant effects the structure has on the propagation. In this work, we are interested in the losses generated by the structure and the material and as such frequency is fixed in order to solve wavenumbers.

1.2.1. Computing the modes of porous layers using semi-analytical tools

The literature on the dispersion relation of porous/poroelastic-based structures is very scarce. To our knowledge, Groby *et al.*⁹ have shown the complex dispersion relation in a homogeneous equivalent fluid layer coupled to air while Boeckx *et al.*¹⁰ have computed the modes propagating in a poroelastic waveguide coupled at each top and bottom interface with an outgoing plane wave propagating in an air medium. This has been done using an analytical description of the system by expressing the interface conditions in a linear system (which we call GM in this work). Both of these work have used a root-finding procedure in order to solve the modes of the structure.

Once the linear system is obtained, we are faced with either or two of these possibilities. Solving wavenumbers can be done through the writing and solving of a (possibly generalized) eigenvalue problem or by root-finding of solutions in the case where writing the first one appears impossible.

Using Eq. (1.6), root finding such that $\det(\mathbf{M}(k_1, \omega)) = 0$; the determinant of the matrix from Eq. (1.5) can be expressed as

$$2 \frac{k_{2,eq} k_{2,0}}{\rho_{eq} \rho_0} \cos(k_{2,eq} h) + i \left(\frac{k_{2,eq}^2}{\rho_{eq}^2} + \frac{k_{2,0}^2}{\rho_0^2} \right) \sin(k_{2,eq} h) = 0. \quad (1.10)$$

From this expression, one can use a root-finding procedure to solve for either the term $k_{2,eq}$ or k_1 to obtain the dispersion relation by varying the frequency. Results are shown in fig. 1.2

Fig. 1.2: Dispersion relation of a rigidly-backed porous layer.

We reference to Appendix ?? for extensive details on the root-finding procedure based on the Müller method. These results show an infinite amount of modes [comment the fig.](#).

Generally, this kind of characteristic equation can only be obtained for very simple cases as is the case here, for a single fluid layer. The characteristic equation for Lamb modes is also well known¹¹. However, more complicated geometry requires to solve the root-finding iterative process by using directly a numerically-computed matrix determinant, or in many case, it is outright not possible due to the analytical complexity of the description of the system. In this case, more complex numerical method are thus required; a discretization of the geometry and an interpolation of the physical fields is done.

1.2.2. A bestiary of the wave solutions in dissipative and open waveguides

In conservative and closed systems, propagating waves are either guided or trapped waves. The total acoustic energy is thus confined to the structure enclosed by its boundary conditions. Nevertheless, in open systems, the acoustic energy is able to radiate outside of the waveguide. Part of propagating waves are thus attenuated and radiate energy to the open domain. They are thus considered quasi-guided since they are propagating waves with some attenuation. Trapped modes can still occur. New family of modes appear as leaky backward and leaky forward waves that exhibit unique properties. Their counterparts are incoming waves.

The addition of dissipation leads to the mixing of several loss mechanisms: the leakage of waves and the dissipations within the guide (which can be of whatever nature). In this case, the separation between incoming and leaky waves is not possible anymore. This particular aspect has been very scarcely tackled in the literature: an unpublished paper is interested in an elastic waveguide¹² while in optics, a paper discusses the waves guided in nanospheres with a complex permittivity¹³.

[manque de contenu](#)

1.2.3. Numerics for modes computation

Regarding the computation of complex modes in homogeneous layer, several numerical methods have been investigated in the last decades. Guided waves represents a large topic and we focus here on the case of simple geometries; similar configurations are considered, an unbounded single (or stack of) layers with two infinite interfaces as presented in the section where we were interested in their acoustic response.

Spectral methods have been around for quite some time and have been used for many numerical applications. We focus for this work on Spectral Collocation Method (SCM)¹⁴, which has been especially used for the computation of guided wave modes in elastic layers^{15–17}. We refer to the introduction of Chapter ?? for an extensive bibliography on SCM applications.

This method is useful for the discretization of 1D geometries as is the object of the present section.

In the case of simple 2D geometries, a 1D discretization along the transverse direction associated to a harmonic motion in the longitudinal direction. Even if the use of finite elements seems farfetched, this method could be relevant in contexts where boundary effects or guide with varying cross-sections are studied. However, a variation of finite elements based on a 1D discretization of the waveguide is entitled Semi-analytical Finite Elements (SAFE) and is akin to SCM which we have described in the previous paragraph. The 1D discretization is done along the thickness of the layer by regular nodes, and the fields are interpolated using finite elements shape functions. Examples of Semi-analytical Finite Elements (SAFE) use cases are very similar to SCM and include [lister des publis ici](#). Examples are also given in Chapter ?. A detailed description of the calculation of Lamb modes using a simple SAFE implementation using high-order Lobatto polynomials is proposed in Appendix. ?. There, its performance is also compared to that of SCM.

scm¹⁵, sort of pml in scm¹⁷ vs local coupling¹⁶ safe has been widely used for this type of problems as well [cite paper with leaky waves and elastic layer + paper SAFE-PML 2.5D](#)

There is a discussion to be made on the relevance of the imposition of a local coupling contrary to a full modelling of the surrounding media with some termination condition satisfying the Sommerfeld condition, PML or analogous absorbing layer. While leading to no assumptions on the radiating field nature contrary to imposing a local coupling, the modeling of the half-space by using some artificial absorbing layer at the end leads to the apparition of modes of the cavities¹⁸.

We argue that the advantages of the imposition of a local coupling is two-fold: less degrees of freedom are necessary since the sole interface is modeled, not the whole half-space. For dispersion problems, this is particularly interesting since we directly obtain leaky mode solutions, without needing to filter out all radiation modes linked to the additional cavity and the resonances of the PML

[relire, réécrire et compléter cette section](#)

gm, tmm, scm, safe, and fem

Fig. 1.3: Discretization for each method

References

1. T. Weisser et al. “Acoustic Behavior of a Rigidly Backed Poroe-lastic Layer with Periodic Resonant Inclusions by a Multiple Scattering Approach”. *J. Acoust. Soc. Am.* 139.2 (2016), 617–629.
2. W. T. Thomson. “Transmission of Elastic Waves through a Strat-ified Solid Medium”. *Journal of Applied Physics* 21.2 (1950), 89–93.
3. N. A. Haskell. “The Dispersion of Surface Waves on Multilay-ered Media*”. *Bulletin of the Seismological Society of America* 43.1 (1953), 17–34.
4. M. Bruneau. *Fundamentals of Acoustics*. London: ISTE, 2006.
5. J.-F. Allard and N. Atalla. *Propagation of Sound in Porous Me-dia: Modelling Sound Absorbing Materials*. 2nd ed. Hoboken, N.J: Wiley, 2009.
6. N. Jiménez, O. Umnova, and J.-P. Groby, eds. *Acoustic Waves in Periodic Structures, Metamaterials, and Porous Media: From Fundamentals to Industrial Applications*. Vol. 143. Topics in Ap-plied Physics. Cham: Springer International Publishing, 2021.
7. O. Doutres, N. Atalla, and H. Osman. “Transfer Matrix Mod-eling and Experimental Validation of Cellular Porous Material with Resonant Inclusions”. *J. Acoust. Soc. Am.* 137.6 (2015), 3502–3513.
8. B. A. Auld. *Acoustic Fields and Waves in Solids. 1*. New York: Wiley, 1973.
9. J.-P. Groby, A. Wirgin, and E. Ogam. “Acoustic Response of a Periodic Distribution of Macroscopic Inclusions within a Rigid Frame Porous Plate”. *Waves Random Complex Media* 18.3 (2008), 409–433.
10. L. Boeckx, P. Leclaire, P. Khurana, C. Glorieux, W. Lauriks, and J. F. Allard. “Investigation of the Phase Velocities of Guided Acoustic Waves in Soft Porous Layers”. *J. Acoust. Soc. Am.* 117.2 (2005), 545–554.
11. D. Royer and T. Valier-Brasier. *Ondes élastiques dans les solides*. Collection Ondes. London: ISTE éditions, 2021.
12. D. A. Kiefer, M. Ponschab, and S. J. Rupitsch. “From Lamb Waves to Quasi-Guided Waves: On the Wave Field and Radia-tion of Elastic and Viscoelastic Plates”. *Unpublished* (2020).
13. S. Campione, S. Steshenko, and F. Capolino. “Complex Bound and Leaky Modes in Chains of Plasmonic Nanospheres”. *Opt. Express* 19.19 (2011), 18345.
14. L. N. Trefethen. *Spectral Methods in MATLAB*. Society for Industrial and Applied Mathematics, 2000.
15. A. T. I. Adamou and R. V. Craster. “Spectral Methods for Mod-elling Guided Waves in Elastic Media”. *J. Acoust. Soc. Am.* 116.3 (2004), 1524–1535.
16. D. A. Kiefer, M. Ponschab, S. J. Rupitsch, and M. Mayle. “Cal-culating the Full Leaky Lamb Wave Spectrum with Exact Fluid Interaction”. *J. Acoust. Soc. Am.* 145.6 (2019), 3341–3350.
17. E. Georgiades, M. J. S. Lowe, and R. V. Craster. “Leaky Wave Characterisation Using Spectral Methods”. *J. Acoust. Soc. Am.* 152.3 (2022), 1487–1497.
18. M. Gallezot, F. Treyssède, and L. Laguerre. “Contribution of Leaky Modes in the Modal Analysis of Unbounded Problems with Perfectly Matched Layers”. *J. Acoust. Soc. Am.* 141.1 (2017), EL16–EL21.